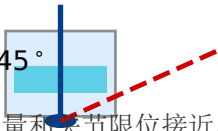




防洒约束: $\theta = \cos^{-1}(\hat{z}_{cup} \cdot \hat{z}_{world}) \leq 45^\circ$

优化目标: 最小化末端误差、关节运动量和关节限位接近度



— 杯体轴线
- - 45° 阈值